

Deep Learning for Space, Time, Graph

Programming Assignment 1

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Code repository: [GitHub - khwrali011/DL4STG_PA1](#) · [GitHub](#)

Question 1

Response 1A

Hypothesis: From lowest to highest expected RMSE:

established stations → period-routed ridge, Raw Attention, sensor-specific ridge, shared ridge

Station 4 → period-routed ridge, Raw Attention, shared ridge.

Period routing explicitly conditions on pace; Attention can adapt to context; station-specific fitting might offer a small benefit from station differences. Sensor-specific ridge cannot serve an unseen station.

Measured ordering.

For established stations, lowest to highest RMSE is period-routed ridge, Raw Attention, shared ridge, then sensor-specific ridge.

For Station 4 the order is period-routed ridge, Raw Attention, then shared ridge.

Relative to the stated hypothesis, the first two positions agree; shared ridge narrowly outperforms sensor-specific ridge rather than the reverse. This small difference does not support a benefit from station-specific fitting.

	Established validation		Station 4 validation	
Model	RMSE	MSE	RMSE	MSE
Shared ridge	0.767	0.588	0.876	0.767
Sensor-specific ridge	0.768	0.590	N/A	N/A
Period-routed ridge	0.166	0.028	0.173	0.030
Raw Attention	0.433	0.188	0.644	0.415

Table 1. Output 1.1. RMSE in m/s^2 and MSE in $(m/s^2)^2$. N/A means no fitted Station 4 model.

	Established validation RMSE			Station 4 validation RMSE		
Model	A	B	C	A	B	C
Shared ridge	.805	.854	.621	.909	.982	.713
Sensor-specific ridge	.804	.852	.630	N/A	N/A	N/A
Period-routed ridge	.181	.164	.152	.184	.182	.150
Raw Attention	.425	.455	.418	.668	.608	.655

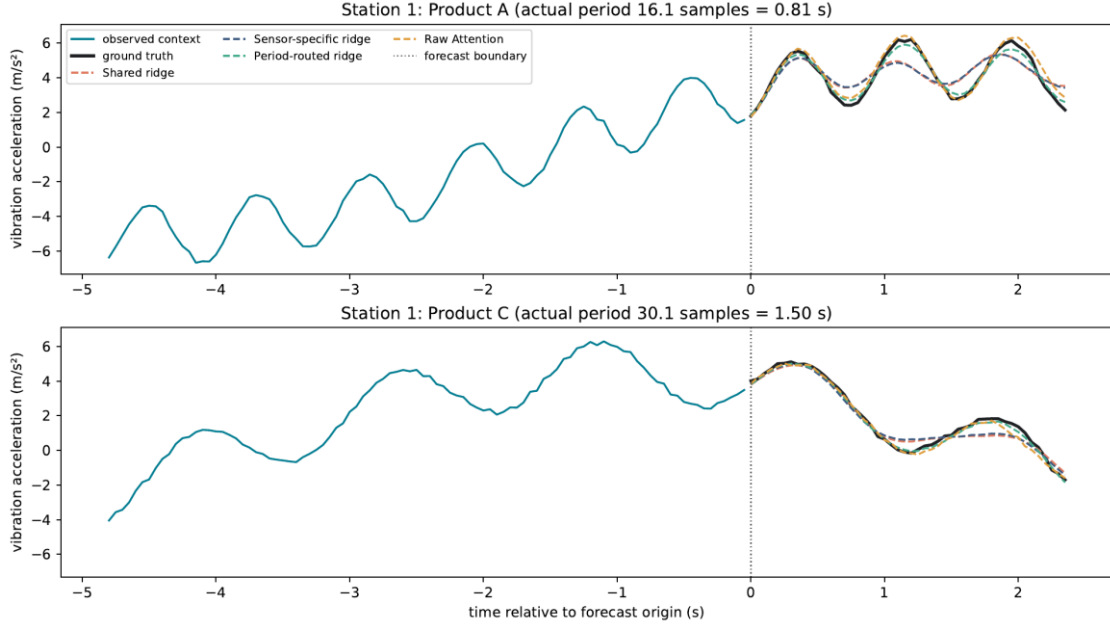
Table 2. Product-level validation RMSE from Output 1.1; N/A denotes an unavailable forecast.

Period routing reduces RMSE relative to shared ridge by 78.4% on established stations and 80.3% on Station 4. Raw Attention reduces it by 43.5% and 26.4%, respectively. Routing is best for every product in both populations. Station-specific fitting barely changes established RMSE (0.767 to 0.768), because station identity does not identify the current pace. Shared and station-specific maps must fit many paces with one operator; period routing supplies the relevant conditioning variable. Attention adapts its mixing to the context but does not receive that explicit routing rule.

Response 1B

The same Station 1 has a period of 16.135 samples in the Product A example and 30.093 samples in Product C. A fixed ridge matrix is reused at both paces. In Output 1.2, its continuation follows the broad level but attenuates the vibration: Product A's later troughs are too shallow and the later peak is too low. Product C's second crest is also flattened. These shape errors show the compromise across different required continuations.

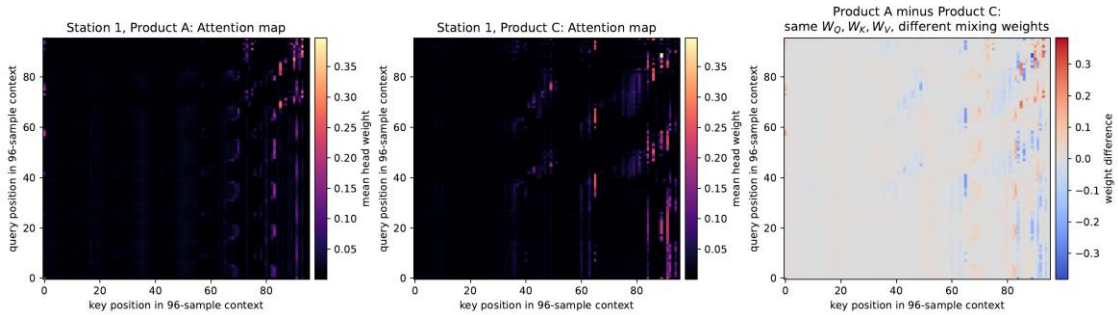
For the displayed Product A window, shared-ridge RMSE is 0.779, compared with 0.232 for period-routed ridge and 0.328 for Raw Attention. In Product C the corresponding values are 0.448, 0.150 and 0.251. These are individual validation windows, not population averages.



Output 1.2: validation forecasts for the same station at two operating paces. Original exported figure.

Attention uses $Q = XW_Q$, $K = XW_K$ and $V = XW_V$, then computes $A(X) = \text{softmax}(QK^\top/\sqrt{d_h})$. The learned projection matrices remain fixed at inference, but changing X changes the pairwise scores, weights and values. Thus the operation $A(X)V(X)$ depends on the current context. Ridge instead reuses the same fitted context-to-future matrix.

In Output 1.3, the mean absolute attention-weight difference between the two contexts is 0.0121 and the maximum is 0.3827. This demonstrates context-dependent mixing; it does not establish recovery of the physical period.

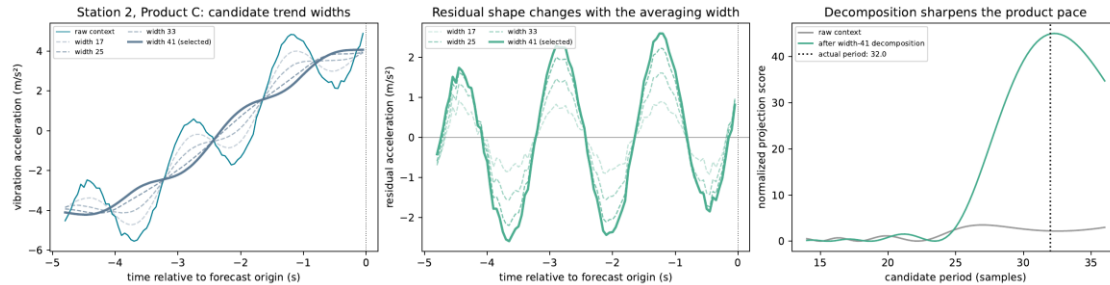


Output 1.3: attention maps and their difference for the displayed validation contexts.

Response 2

Implementation: A centered moving average with odd width m estimates the trend T . Padding repeats the nearest endpoint; the remainder is $S = x - T$. Both outputs retain the input shape. For $m = 3$ and $x = [1, 2, 3, 10, 5]$, the trend is $[4/3, 2, 5, 6, 20/3]$. The supplied value, reconstruction and gradient checks passed.

Output 2.1 shows that narrower windows follow more of the oscillation, leaving a smaller residual. As width rises from 17 to 41, the displayed residual standard deviation rises from 0.578 to 1.572 m/s². The wider trend smooths over more of the vibration. Repeated endpoint values affect the edge estimates; the residual is an estimated remainder, not the generator’s true repeating component.



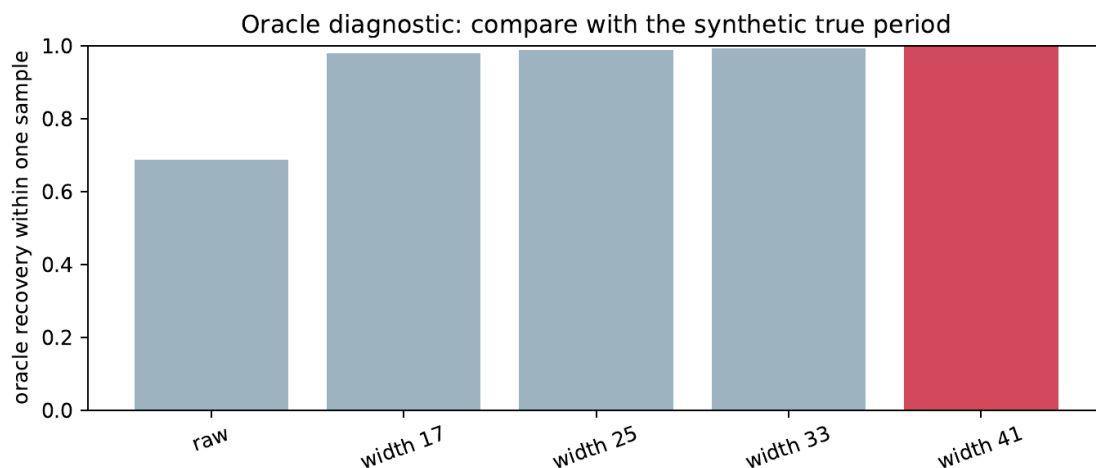
Output 2.1: width-dependent trend, residual and period evidence for one validation context.

Representation	Recovery within one sample	Median absolute period error
Raw	68.67%	0.525
Width 17	98.00%	0.157
Width 25	98.92%	0.122
Width 33	99.33%	0.142
Width 41	99.85%	0.216

Table 3. Output 2.2. Training-only oracle period diagnostic; errors in samples.

Selected width: 41 samples.

Width 41 is selected by the highest fraction of period estimates within one sample, not by the lowest median error (which belongs to width 25). Access to the synthetic true period makes this diagnostic possible; it is not an input to the neural forecaster and would not ordinarily be available in deployment.



Output 2.2: oracle recovery criterion used for width selection.

A moving average is suitable here because the 240-sample drift is slower than the 14–36-sample vibration. Separating them leaves repetition for the encoder and a smooth component for the linear trend head. On validation, adding decomposition to Attention reduces established MSE from 0.188 to 0.083 (55.6%) and Station 4 MSE from 0.415 to 0.183 (55.9%); RMSE becomes 0.289 and 0.428. All product-level validation RMSEs improve (Output 2.3). An abrupt baseline jump would violate the smooth-trend assumption: averaging would spread the jump over a window and leave a large residual near it.

Response 3

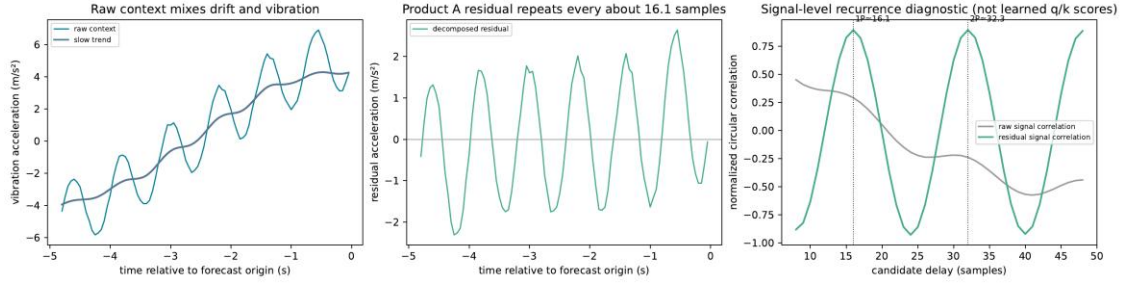
Numerical verification. The supplied scorer centers $[1,3,1,3]$ to $[-1,1,-1,1]$. Direct circular correlation and the FFT both give $[4,-4,4,-4]$ for delays 0–3. The implemented aggregation uses

$$z_t = \sum_j \alpha_j v_{(t-\tau_j) \bmod L}.$$

With values $[10,20,30,40]$, delays $[1,3]$ and weights $[3/4,1/4]$, the output is $[35,15,25,25]$. In particular,

$$z_0 = (3/4)40 + (1/4)20 = 35,$$

confirming the shift direction. Each example has its own delays and weights, shared across heads and features. The supplied aggregation and gradient checks passed.



Output 3.2: direct signal-level circular correlations for a validation context. These are not learned query/key scores.

Recurrence evidence:

- ➔ For Station 2's displayed Product A run, $P = 16.135$ samples. The residual has strong peaks near 16 and 32 samples, corresponding to P and $2P$.
- ➔ The reported correlations at those integer lags are 0.896 and 0.897. A further peak appears near the upper end of the plotted delay band, close to $3P \approx 48.4$ samples.

The raw signal contains a large smooth rise, so its lag correlation is influenced strongly by drift. Removing the slow part exposes alternating alignment and misalignment of the repeating vibration. This supports grouping relationships by delay: positions separated by approximately one cycle are related for the same reason. The model itself applies delay scoring to learned query; this figure illustrates the signal assumption only.

Failure case:

- ➔ An isolated impact is not a recurring cycle. Averaging shifted copies can spread the impact to unrelated positions or wrap it across the context boundary, creating misleading evidence for repetition.
- ➔ Similarly, a period that changes rapidly within the window cannot be described well by one small set of global delays. Content-dependent pairwise attention can retain distinctions that such shared delays discard.

Boundary assumption: Circular indexing treats the last and first context positions as adjacent. This makes all delays use the same number of products, but the artificial join can distort scores when the context does not contain an integer number of cycles.

Response 4

Hypothesis:

- Decomposition should be especially helpful when slow drift is prominent because it separates the baseline from the repeating vibration.
- Delay mixing should also help by grouping recurring displacements, but the combined gains may overlap because both changes address the same drift-plus-repetition structure.

	Established val.		Station 4 val.			
Model	RMSE	MSE	RMSE	MSE	Params	Fit (s)
Shared ridge	.767	.588	.876	.767	4,608	.003
Sensor-specific ridge	.768	.590	N/A	N/A	13,824	.005
Period-routed ridge	.166	.028	.173	.030	59,904	.143
Raw Attention	.433	.188	.644	.415	368,368	283.8
Attention + decomposition	.289	.083	.428	.183	373,024	416.4
Raw delay mixer	.224	.050	.353	.125	368,370	372.8
Autoformer-inspired	.197	.039	.243	.059	373,026	509.0

Table 4. Outputs 1.1 and 4.1. Validation accuracy and fitting cost; N/A is unavailable.

Both switches improve aggregate validation error in either configuration of the other switch.

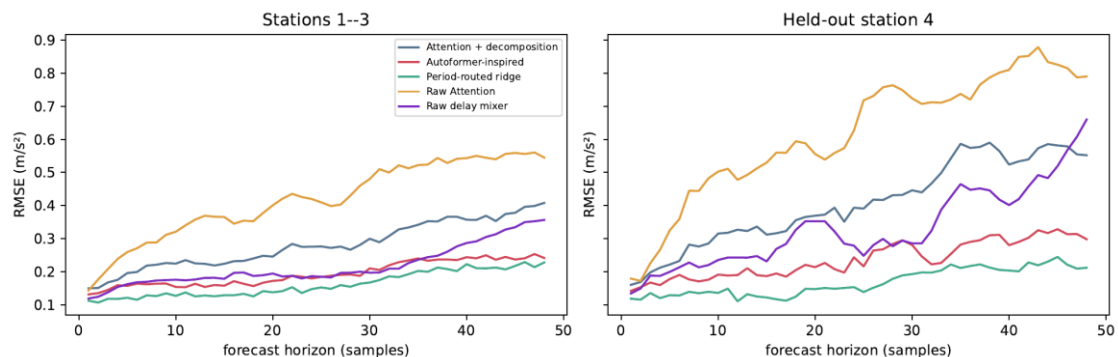
→ On established stations, decomposition reduces MSE by 0.1042 with Attention but only 0.0113 with delay mixing. Replacing Attention with delay mixing reduces MSE by 0.1375 without decomposition and 0.0446 with it. The combined reduction (0.1488) is smaller than the sum of the standalone reductions (0.2417).

→ Station 4 shows the same subadditive pattern: 0.3559 versus 0.5225. This describes overlapping gains in this run; it does not establish their cause.

Errors generally increase with horizon, with local fluctuations. Established early/late RMSE (steps 1–12/37–48) is 0.274/0.547 for Raw Attention, 0.155/0.241 for Autoformer-inspired and 0.123/0.215 for period-routed ridge. On Station 4 these are 0.386/0.817, 0.174/0.307 and 0.130/0.218.

Selection: period-routed ridge for both populations:

It has the lowest validation error, 59,904 parameters versus 373,026 for Autoformer-inspired, and a 0.143-second fit. Final test RMSE is 0.161 for established stations and 0.183 for Station 4. Autoformer-inspired is the best neural model overall, but not on every individual window.

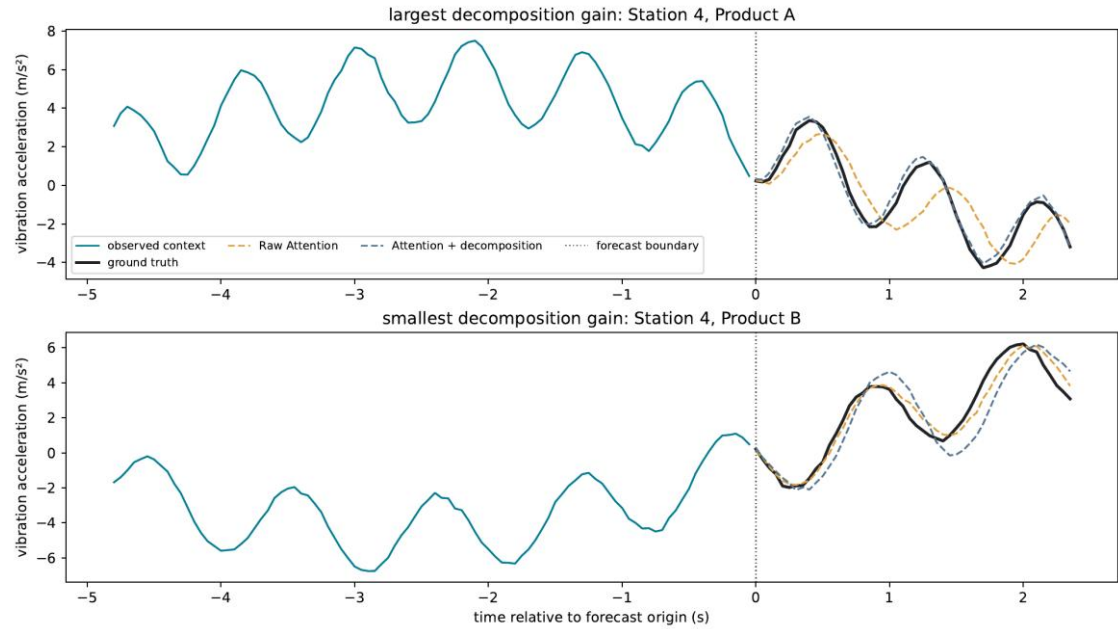


Output 4.2: validation RMSE across the 48-step horizon. Early/late summaries above are square roots of mean per-step MSE, not means of per-step RMSE.

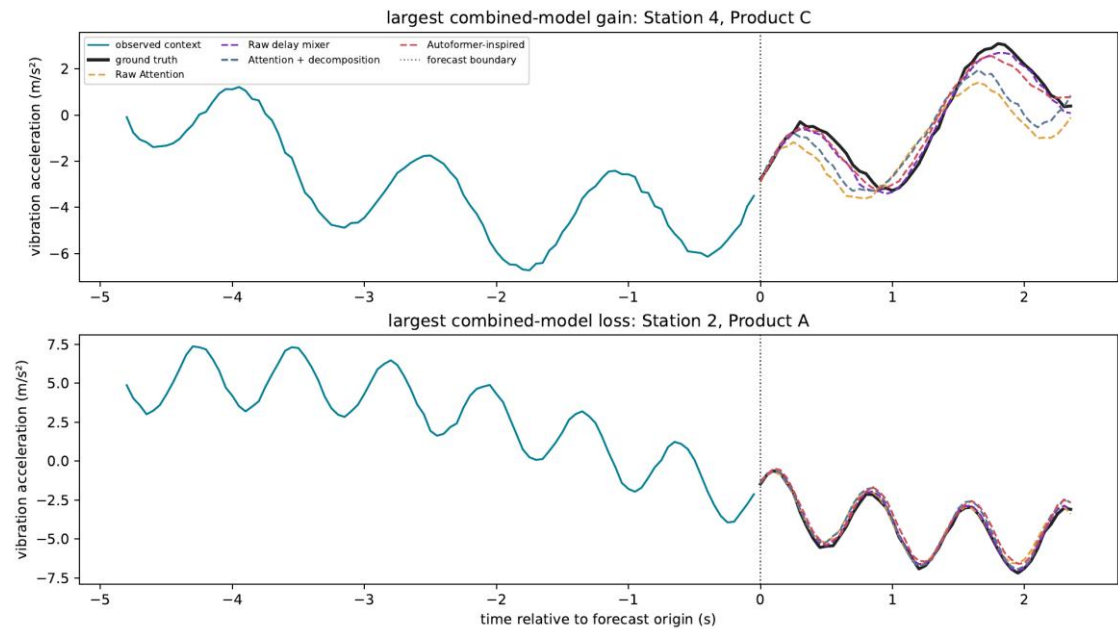
Population	Selected model	Val. MSE	Val. RMSE	Test MSE	Test RMSE
Established	Period-routed ridge	.0275	.1659	.0260	.1611
Station 4	Period-routed ridge	.0298	.1727	.0333	.1826

Table 5. Output 4.3. Final evaluation after validation-based model selection.

Supporting forecast evidence



Output 2.3: contrasting validation examples for adding decomposition to Attention. The first case corrects much of Raw Attention’s later phase mismatch; the second still contains local forecast errors.



Output 4.1: selected validation cases showing the largest combined-model gain and loss relative to Raw Attention. These deliberately contrasting examples do not represent average performance.

Generative-AI use. ChatGPT/Codex was used for implementation assistance, inspection of saved outputs and environment setup guidance. Training and the reported numerical evidence came from the executed notebook on personal system.

Question 2

Attempt 1

Goal and data:

1. The target has 43,656 observed values at unit-spaced integer indices.
2. The task is a direct forecast of the 168 hidden values at indices 43,657–43,824.
3. The target is non-negative and heavy-tailed; no calendar frequency is disclosed.
4. Ten optional variables cover all 43,824 indices, including the forecast horizon, and their future values are explicitly permitted by the handout.
5. Hidden target values were never used for fitting.

Architecture: The architecture used in both submissions is a compact Autoformer.

- ➔ A centered 25-step moving average separates seasonal and trend components; decomposition also follows each encoder and decoder mixing/feed-forward operation.
- ➔ Four-head query/key features are scored by FFT cross-correlation over delays.
- ➔ The five highest-scoring delays aggregate shifted value features, with scores shared over heads/features for each example.
- ➔ There is one encoder layer and one decoder layer, 32 hidden channels and a 64-channel feed-forward sublayer.
- ➔ The decoder sees the latest 168 seasonal observations, zero placeholders for the next 168, a trend continuation initialized from the observed trend mean, and optional variables for its entire 336-step decoder input.
- ➔ The encoder uses 672 observed target/optional steps; cross-correlation uses its latest 336 states. This retains decomposition and delay-level aggregation, the two characteristic Autoformer mechanisms, although it is a compact implementation rather than the published configuration.

Processing and objective. The target is transformed by $\log(1 + y)$ and standardized using the training portion only. Six continuous optional columns are standardized on the same prefix; sparse continuous columns E and F first receive $\log(1 + x)$. The four indicator columns remain binary. All target predictions are converted back to non-negative original units before scoring. The selected training objective is

$$\mathcal{L} = 0.7 \text{MSE}\left(\frac{\hat{y}}{100}, \frac{y}{100}\right) + 0.3 \text{MSE}(\hat{z}, z), \quad z = \frac{\log(1 + y) - \mu_{\text{train}}}{\sigma_{\text{train}}}.$$

Training uses

- ➔ AdamW (learning rate 5×10^{-4} , weight decay 10^{-4}), batch size 32, window-origin stride 16, gradient clipping at 1, at most 12 epochs and patience 4.

Chronological validation:

- ➔ Fold A fits/scales on indices 1–27,000 and scores nine complete, non-overlapping 168-step blocks beginning at index 27,001, spaced 672 apart, before index 33,000.
- ➔ Fold B fits/scales on indices 1–33,000 and scores sixteen analogous blocks beginning at index 33,001, ending before the withheld continuation. Observed values before each validation origin may be used as its context, but validation targets are never fitted or included in scale estimation.

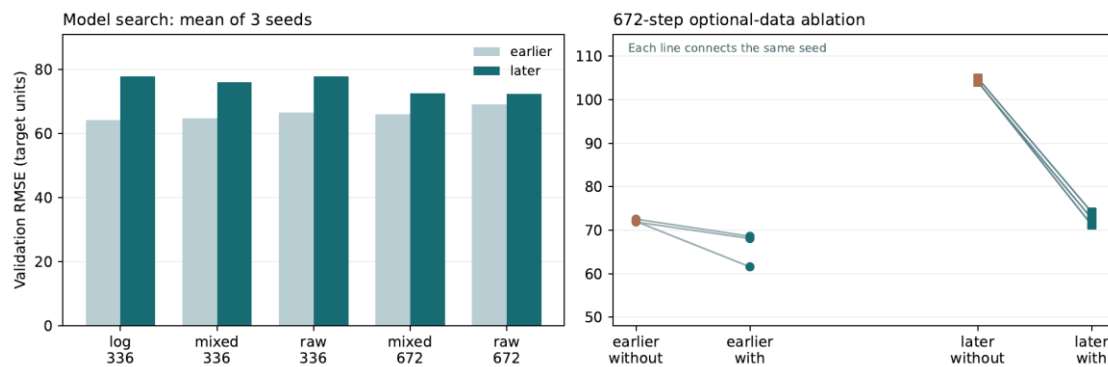
The same folds and seeds 17,29,43 are used across five configurations. Scores in the next table are arithmetic means of each seed's 168-step RMSE; the overall column weights the two folds equally, rather than weighting each individual forecast point equally.

Objective / context	Earlier fold	Later fold	Two-fold mean
Log MSE / 336	64.34	77.92	71.13
Mixed / 336	64.74	76.02	70.38
Raw MSE / 336	66.60	77.88	72.24
Mixed / 672	66.09	72.65	69.37
Raw MSE / 672	69.18	72.44	70.81

Table 6. Question 2 model search. RMSE in anonymised target units, mean across three seeds.

Initial model selection:

- ➔ The longer context helps the mixed-loss model on the later fold (76.02 to 72.65 RMSE) but hurts it on the earlier fold (64.74 to 66.09).
- ➔ Mixed loss with 672-step context has the best equal-fold mean, 69.37.
- ➔ Seed 29 has the lowest two-fold mean within that configuration (67.85), so it was selected for the first full-history refit.
- ➔ The one-fold prototype had validation RMSE 80.11 with optional data; its population matches Fold B, where the revised configuration scores 72.65 averaged over seeds. These figures show validation improvement; actual hidden-test results appear below.



Question 2 validation evidence. Left: search RMSE averaged over three seeds. Right: the optional-data ablation for the selected 672-step configuration; each line pairs the same seed within one fold. Source: exported search and ablation results.

Optional variables and final forecast

Matched optional-data ablation. Run held the 672-step model, mixed loss, window construction, folds and three seeds fixed, changing only whether the ten optional columns enter the model. The six matching “with” runs were recovered from the search; six “without” runs were newly trained under the same settings. Each arm selected its best epoch on the same validation blocks, as in the search. The optional variables improve RMSE in all six seed/fold pairs; the advantage is much larger in the later fold. MAE and sMAPE also improve in the equal-fold comparison.

Validation population	Without optional	With optional	Relative RMSE reduction
Earlier fold (9 blocks; 3 seeds)	72.15	66.09	8.4%
Later fold (16 blocks; 3 seeds)	104.38	72.65	30.4%
Equal-fold mean	88.27	69.37	21.4%

Table 7. Matched optional-data ablation. Mean per-run 168-step RMSE across three seeds.

→ Equal-fold MAE changes from 62.22 without the optional file to 45.95 with it;

→ sMAPE changes from 68.82% to 50.44%.

This supports feeding the measured covariates into the encoder and forecast-horizon decoder for this dataset. The experiment does not isolate individual features or prove that all ten are useful. Future optional measurements are provided by the assignment and are explicitly allowed; no future target values are used.

Attempt 1 fitting and cost. *The selected seed 29 model was refitted on all 43,656 observed target values*, with preprocessing recomputed on that history. Its two validation fits selected epochs 2 and 4; their median gave a three-epoch final refit. The model has 22,210 trainable parameters. The declared 17 epochs comprise 14 executed epochs in the selected seed's two validation runs and three in the final refit. The broader initial search used 259 validation-training epochs across 30 runs; that exploratory cost is reported separately from the submitted model's selected pipeline.

Results:

The first actual leaderboard submission returned RMSE 106.2365, MAE 70.3895 and sMAPE 62.58%. The higher test RMSE than the 69.37 validation mean motivated a closer examination of recent chronological windows.

Attempt 2

The follow-up compared five candidates across seeds 17, 29 and 43:

- ➔ mixed/672 with full history
- ➔ raw/672 with full history
- ➔ mixed/672 with the latest 16,000 observations
- ➔ raw/336 with the latest 16,000 observations
- ➔ and mixed/336 with full history

All used the same compact Autoformer family and optional variables. The follow-up separated early stopping, development selection and confirmation into consecutive 168-step blocks, so the confirmation targets did not determine the chosen candidate, seed or epoch count.

Each candidate first chose its best epoch on the early-stopping region. It was then refitted on the development prefix for that fixed number of epochs and scored on the next four blocks. The configuration was selected by development RMSE averaged over three seeds; the seed with the lowest development RMSE within the winning configuration was selected for the final pipeline. The chosen model was refitted on the confirmation prefix for diagnostic evaluation, then fitted afresh on the complete observed training history.

Stage	Fit prefix	Scored target indices	Purpose
Early stopping	1–41640	41641–42312	Choose epoch count
Development	1–42312	42313–42984	Choose configuration and seed
Confirmation	1–42984	42985–43656	Check fixed selection
Final fit	1–43656	43657–43824	Generate hidden-test forecast

Table 8. Recent-window protocol. All ranges are inclusive; each scored region contains four non-overlapping 168-step blocks. Hidden-test targets are not available locally.

Training and selection:

1. The follow-up retained AdamW with learning rate 0.0005, weight decay 0.0001, batch size 32, stride 16 and gradient clipping at 1.
2. It allowed at most 12 early stopping epochs, with patience 3.
3. The selected candidate remained mixed/672 with full history and all optional variables, but the selected seed changed from 29 to 43, and the fixed refit length increased from three to six epochs.
4. The winning configuration's development RMSE averaged over seeds was 69.9045. This is a different validation population from the initial 69.37 two-fold mean, so the numbers do not establish a direct improvement between searches.

Confirmation results:

- ➔ The selected seed 43 model obtained RMSE 61.9479, MAE 34.7305 and sMAPE 55.26% on the final four known training blocks.
- ➔ Their individual RMSEs were 57.88, 97.61, 37.70 and 32.42, showing considerable variation across windows.
- ➔ On the same confirmation population, repeating the last observed value gave RMSE 245.17 and repeating the last 168 observations gave RMSE 164.54.

Attempt 2 fitting and cost. The final seed 43 model was refitted for six epochs on all 43,656 observed target values. It still has 22,210 trainable parameters. The declared 27 epochs include the selected seed's early stopping, development, confirmation and final-fit runs.

Question 2 Leaderboard results through Attempt 2

Attempt	Test RMSE	Test MAE	sMAPE	P	E	Score
1	106.2365	70.3895	62.58%	22,210	17	Not recorded
2	91.7374	58.6755	52.44%	22,210	27	91.7510

Table 9. Actual hidden-test results reported by the leaderboard. Score for Attempt 1 was not recorded in the supplied evidence. P is trainable parameters; E is the declared selected-pipeline training epochs.

Results:

- ➔ Attempt 2 reduced actual test RMSE by 14.4991 units (13.65%)
- ➔ MAE by 11.7140 units (16.64%)
- ➔ sMAPE decreased by 10.14 percentage points

Interpretation:

1. Recent-window validation improved the selection protocol, but its confirmation RMSE of 61.95 was still optimistic relative to the actual test RMSE of 91.74.
2. Large changes in a heavy-tailed series and the variation across 168-step blocks limit what one confirmation window can establish.
3. The evidence does not reveal the cause of the hidden-test error because the hidden target sequence is unavailable.

Attempt 3

The comprehensive search selected a larger Autoformer ensemble. Its actual leaderboard test RMSE was 86.4685, improving on Attempt 2's 91.7374. MAE and sMAPE increased, so the improvement was specific to RMSE and the leaderboard score.

Training and selection:

1. The selected configuration used a 672-step context, 64 hidden channels, two encoder layers, a 25-step moving average and normalised correlation scores. It used the raw target and raw MSE objective, all optional variables and an additional covariate prediction head.
2. The three ensemble members used seeds 17, 29 and 43. Their predictions were averaged with equal weights of one third. All three used full training history, learning rate 0.0005 and dropout 0.1.
3. Candidates were screened and refined using three chronological folds. Each fold had four 168-step blocks for early stopping and twelve separate blocks for development scoring. The selection score was $0.75 \times \text{pooled development RMSE} + 0.25 \times \text{worst-fold RMSE}$, with fewer parameters preferred within 1% of the best score.
4. For each selected seed, the final epoch count was the median of its best epochs across the three folds. Model choice, ensemble weights, calibration and epoch counts were fixed before audit evaluation.

Stage	Fit prefix	Early stopping	Development / audit
Fold 1	1–30216	30217–30888	30889–32904
Fold 2	1–35592	35593–36264	36265–38280
Fold 3	1–39624	39625–40296	40297–42312
Audit	1–42312	Fixed epoch count	42313–43656
Final fit	1–43656	Fixed epoch count	Test: 43657–43824

Table 10. Attempt 3 chronological protocol. All target ranges are inclusive. Hidden-test targets were not available for local training or scoring.

Validation results:

- The selected ensemble obtained pooled development RMSE 57.8187, MAE 39.9668 and sMAPE 51.87%. The three fold RMSEs were 37.98, 71.85 and 58.52, showing variation across chronological regions.
- An affine correction was rejected because its chronological cross-check increased RMSE from 65.52 to 68.12. The final ensemble used the uncorrected average.
- On eight audit blocks, RMSE was 64.8406, MAE 51.1232 and sMAPE 70.63%. Audit RMSE rose from 38.15 at steps 1–24 to 62.93 at steps 25–72 and 70.81 at steps 73–168.

The audit region was reserved from selection within this search, although parts had been examined in earlier experiments. Attempt 2's confirmation and Attempt 3's audit covered different populations, so their validation numbers do not establish a direct improvement.

Attempt 3 fitting and cost

The final ensemble was fitted on all 43,656 observed target values. Seeds 17, 29 and 43 were fitted for 8, 4 and 3 epochs respectively, giving 15 final-fit epochs in total. The ensemble has 515,142 trainable parameters across its three members.

The declared 112 epochs include the selected members' three-fold early-stopping runs, audit fits and final fits. The complete search and ablation used 348 executed epochs. Recorded fitting time was 120.65 minutes, approximately two hours. These broader search costs are separate from the selected pipeline's declaration.

Question 2 Leaderboard results through Attempt 3

Attempt	Test RMSE	Test MAE	sMAPE	P	E
1	106.2365	70.3895	62.58%	22,210	17
2	91.7374	58.6755	52.44%	22,210	27
3	86.4685	59.7808	58.02%	515,142	112

Table 11. Actual hidden-test results reported by the leaderboard. *P* is the total trainable parameter count; *E* is the declared selected-pipeline training epochs. Lower values are better for each error metric and score.

Results:

- Attempt 3 reduced test RMSE by 5.2689 units (5.74%) relative to Attempt 2.
- MAE increased by 1.1053 units (1.88%), and sMAPE increased by 5.58 percentage points.
- The leaderboard score improved from 91.7510 to 86.5255 despite the larger parameter count and higher declared epoch count. The recorded snapshot showed rank 29 and three of five attempts used.

Interpretation:

1. Attempt 3 is the best recorded submission by RMSE and leaderboard score. Attempt 2 still has the lowest MAE and sMAPE among the three submissions.
2. The pattern is consistent with a reduction in larger errors accompanied by worse average absolute and relative errors. The hidden target sequence is unavailable, so the affected points cannot be identified.
3. The ensemble uses approximately 23.2 times as many parameters as Attempt 2. Its lower RMSE therefore comes with substantially greater model and training cost.

Submission artifact. The Attempt 3 forecast was generated for time_idx 43657–43824. Preserve question2_comprehensive_output/leaderboard_values.txt with the forecast, declaration, confirmation report and the three final model checkpoints. The submitted declarations were $P = 515,142$ and $E = 112$.

Generative-AI record. ChatGPT/Codex helped in generation of the scripts and inspection of saved outputs.

References:

1 H. Wu et al., “Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting,” *Advances in Neural Information Processing Systems*, 2021.